Earth and Planetary Science Letters, 55 (1981) 53–64
Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

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Preliminary paleomagnetic poles and correlation of the Proterozoic Uinta Mountain Group, Utah and Colorado

Stephen L. Bressler

U.S. Geological Survey, Flagstaff, AZ 86001 (U.S.A.)

Received February 26, 1981
Revised version received April 8, 1981

Stable paleomagnetic directions have been obtained from seven sites spanning much of the 7-km-thick middle Proterozoic Uinta Mountain Group. The characteristic magnetization is carried partly by detrital(?) and secondary specularite and partly by hematite pigment. Preliminary paleomagnetic poles for six of the seven sites are closely grouped and their mean pole is 0.6°N, 157.2°E ($N=6$, $\alpha_{95}=5.4°$). The pole for the seventh and stratigraphically highest site lies to the west, 4.6°N, 140.1°E, and ($n=8$, $\alpha_{95}=15.9°$). On the basis of the poles, revisions to correlations of the Uinta Mountain Group with other sequences are proposed. The Uinta Mountain Group appears to correlate with the Chuar Group and perhaps the upper member of the underlying Nankoweap Formation, and post-dates the Unkar Group of the Grand Canyon Supergroup. In addition, the upper part of the Uinta Mountain Group appears to correlate with the Little Dal Group of the Northwest Territories, Canada. The Uinta Mountain Group appears to entirely post-date the Belt Supergroup of Montana and Idaho. Correlation of the Uinta Mountain Group with the Chuar and Little Dal Groups is supported by the presence of the fossil *Chuaria* and by apparently overlapping isotopic ages. A maximum age range for the Uinta Mountain Group, extrapolated from the Grand Canyon Supergroup and the Little Dal Group, extends from <1100 to >770 m.y.

1. Introduction

This report summarizes preliminary results from a paleomagnetic study being conducted in the middle Proterozoic Uinta Mountain Group of northern Utah and northwesternmost Colorado (Fig. 1). The goal is to establish an apparent polar wandering path and a polarity zonation for correlation of the Uinta Mountain Group with other North American Proterozoic sequences. Results are presented for seven sites collected at widely spaced stratigraphic intervals across a clastic sedimentary sequence that is about 7 km thick. Poles calculated from paleomagnetic directions provide evidence for a correlation with some middle Proterozoic strata of northern Arizona and northwestern Canada.

A paleomagnetic pole has been reported by Runcorn [1] for the Uinta Mountain Group, but samples from the one site he studied were not subjected to demagnetization analysis.

2. Geologic setting

The Uinta Mountain Group constitutes the core of the east-west-trending Uinta Mountains, a broad anticlinal structure formed in Late Cretaceous and early Tertiary time. The Uinta Mountain Group, exposed in a range that is more than 200 km long and about 40 km wide (Fig. 1), is about 7300 m thick in the eastern Uinta Mountains [2] and about 4000 m thick at the western end of the range [3]. The group is undivided in the east and consists mainly of massive-weathering red siliceous sandstone and quartzite that contains minor interbedded conglomerate and shale [2]. In the western Uinta Mountains, the lower part, dominantly quartzite, has been divided into six informal units by Wallace [4], and the quartzite is overlain by the Red Pine Shale. Fig. 2 shows generalized columnar sections of the eastern and western sequences and the approximate stratigraphic positions of sample sites. The locations of sample sites, the strati-

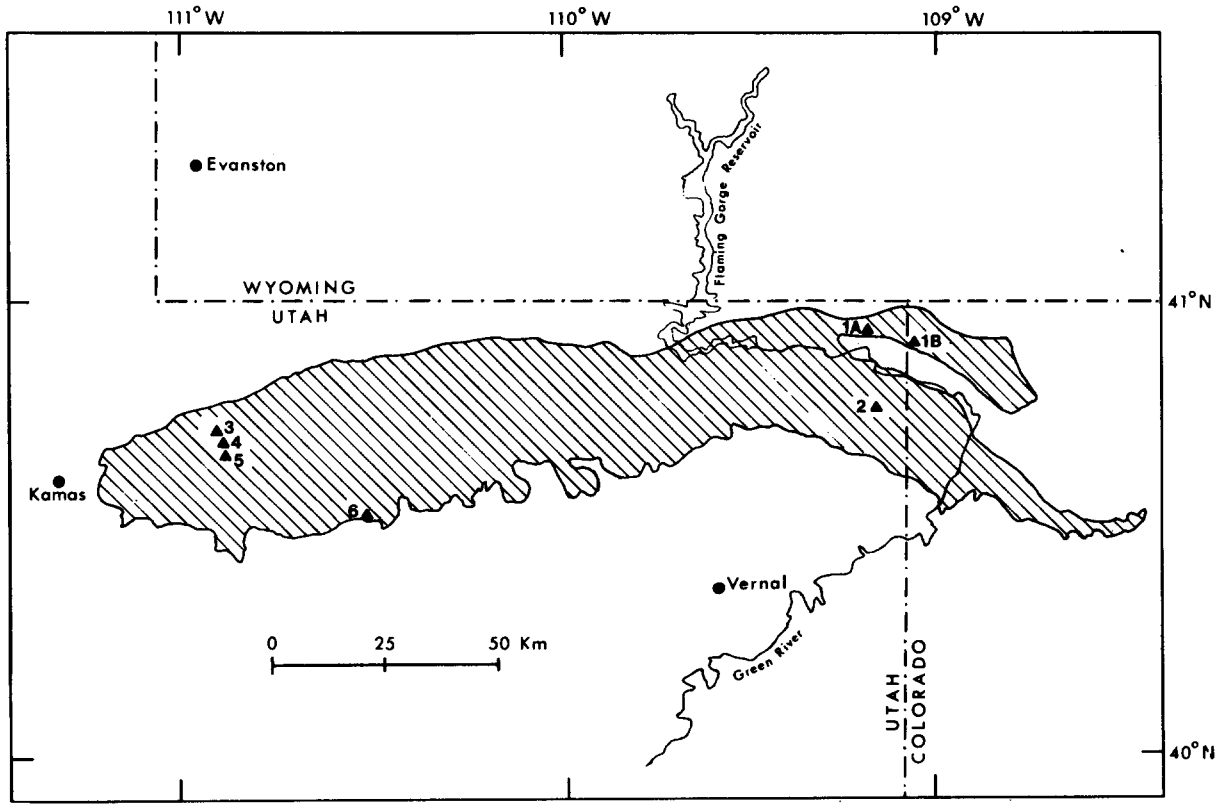


Fig. 1. Index map showing area of outcrop of Uinta Mountain Group (shaded area). Sample site identifications (triangles) represent stratigraphic units and localities as follows: *1A* = lower part of Uinta Mountain Group, Jesse Ewing Canyon; *1B* = lower part of Uinta Mountain Group, Cold Spring Mountain; *2* = middle(?) part of Uinta Mountain Group, Crouse Creek; *3* = shale of Moosehorn Lake of Wallace [4], Bald Mountain Pass; *4* = quartzite of Mount Watson of Wallace [4], Bear River; *5* = quartzite of Hades Pass of Wallace [4], Bear River; and *6* = quartzite of Hades Pass [4], Moon Lake.

graphic units, and brief lithologic descriptions are given in Table 1.

Wallace [6] notes that much of the hematite that coats quartz and feldspar grains in sandstone and quartzite of the western Uinta Mountain Group was formed early, before later diagenetic alterations. Hansen [2] reports that hematite pigment forms rims around many grains upon which quartz overgrowths have subsequently formed. In addition to the apparent early hematite pigment, the quartzite also contains specular hematite. Some of it as seen under the reflecting microscope appears to be of detrital origin, but some of it clearly is secondary.

a Rb-Sr age on muscovite of ~ 2300 m.y. as well as several K-Ar ages of ~ 1500 m.y. were obtained from the Red Creek Quartzite, which underlies the eastern Uinta Mountain Group [2]. Hansen [2] suggests that the ~ 1500-m.y. age represents a thermal event accompanying metamorphism and emplacement of numerous pegmatite dikes. A similar maximum age of 1600 to 1400 m.y. comes from lead model ages obtained from fresh feldspar clasts in arkose of the Uinta Mountain Group in the western part of the range [4].

Crittenden and Peterman [7] report a provisional Rb-Sr whole-rock isochron age of 952 ± 5 m.y. form fine-grained samples of the Red Pine Shale, and they suggest that the age range for the group is about 1400 to 950 m.y. Additional Rb-Sr whole-rock ages from the Uinta Mountain Group that ragne from 1100- to 925-m.y. have been re-

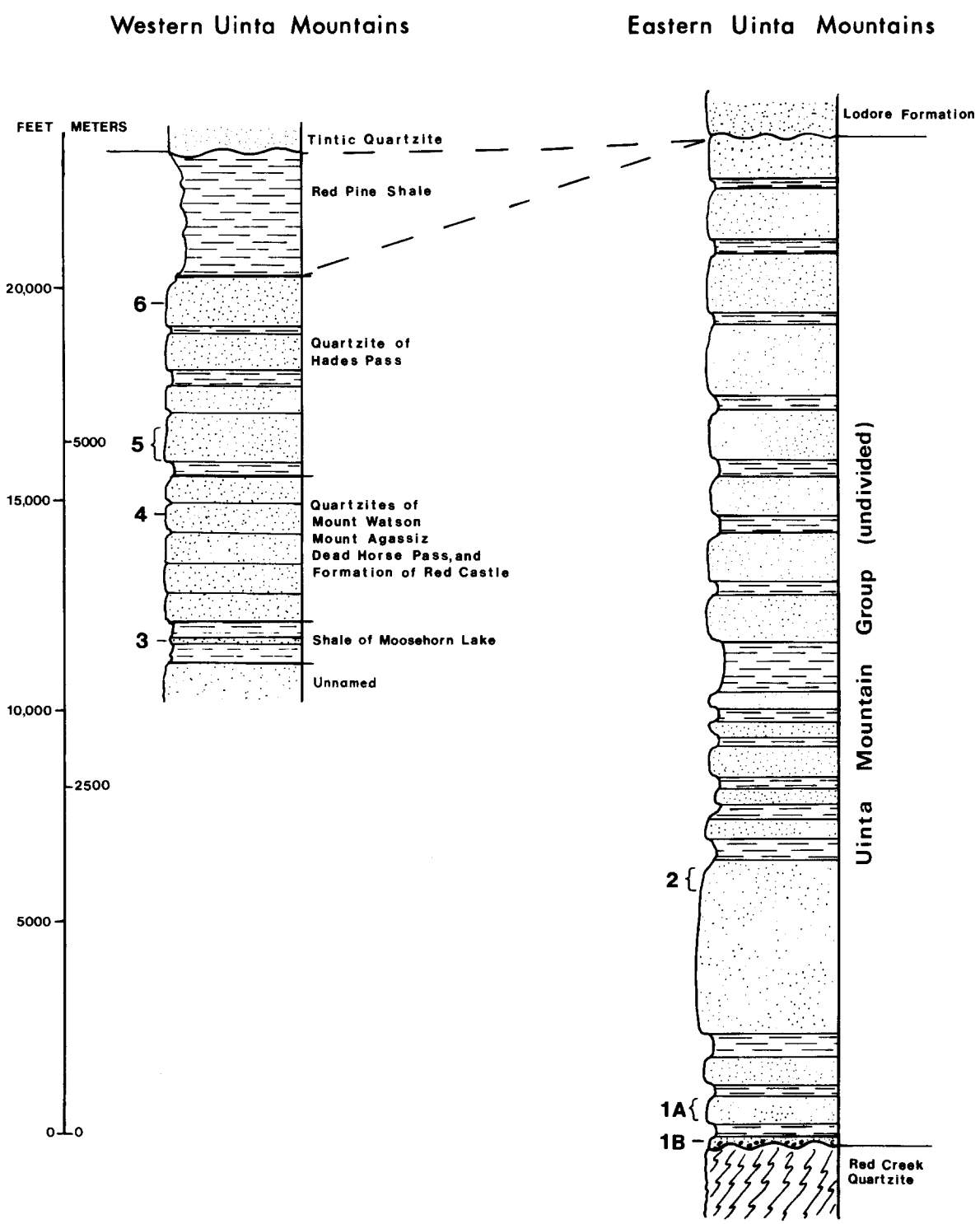


Fig. 2. Generalized stratigraphic sections of western and eastern Uinta Mountain Group, showing approximate stratigraphic position of sample sites. Eastern section from Hansen [2]; western section from Wallace [4]. Informal units beneath Red Pine Shale in western Uinta Mountains are those of Wallace [4]. Site abbreviations as in Fig. 1.

cently reported by Chaudhuri and Hansen [8]. Crittenden and Peterman [7] conclude that the Uinta Mountain Group correlates with the Belt Supergroup of Montana, the age range of which has been reported as 1450 to 850 m.y. [9]. An age range of 1400 to 925 m.y. for the Uinta Mountain Group implies a correlation with parts of other middle Proterozoic sequences of western North America, including the Grand Canyon Supergroup ($\sim$ 1300 to $\sim$ 825 m.y.) and the Apache Group and related rocks of Arizona [10].

Paleontologic age information for the Uinta Mountain Group consists of the problematic fossil *Chuaria* in the Red Pine Shale [11], a small disk-shaped acritarch that has been found in at least nine other localities elsewhere in the world. The age range of *Chuaria* is reported to lie in the interval from 1100 to 600 m.y. ago [11]. The age of *Chuaria* in the Red Pine Shale, 925 to 950 m.y. on Rb-Sr analysis, lies in the early part of the reported age range.

3. Field and laboratory procedures

Oriented core samples 5–10 cm long and 2.5 cm in diameter were obtained using a portable gasoline-powered diamond core drill [12]. Orientations were measured with a magnetic compass and clinometer, read to the nearest one-half degree. Samples were collected form the freshest exposures available, chiefly road cuts and stream courses. Total thickness of strata sampled at each site averaged 20 m, ranging from 3 m at site 3 to 300 m at site 5. Stratigraphic separations within sites were typically one-third meter. However, at site number 1 (Figs. 1 and 2), spacings were about 10 m.

After the specimens were trimmed, the natural remanent magnetization (NRM) of the specimens was measured on a superconducting rock magnetometer having a background noise of $< 10^{-7}$ emu. All samples were then thermally demagnetized in eight steps from 250 to 670°C to determine intervals of stable magnetization. Thermal demagnetization analysis was employed because hematite, with its high magnetic coercivity, was the only magnetic mineral observed in the specimens.

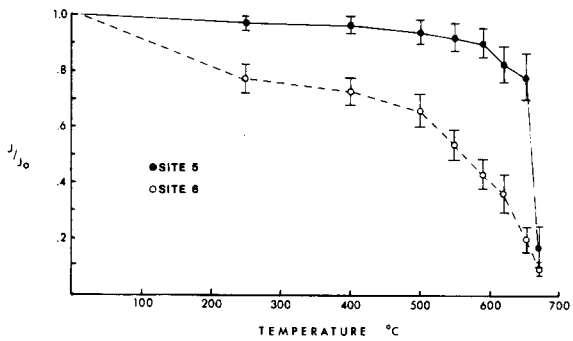


Fig. 3. Relative intensity of magnetization versus demagnetization temperature for all samples from two sites. Error bars represent ± 1 standard deviation.

Stable magnetizations, presumably acquired early in the rock's history, were obtained from samples from the seven sites shown in Figs. 1 and 2. Samples from two additional sites contain magnetizations directed along the present axial field that failed to respond to demagnetization. The rock at these two sites is medium- to coarse-grained, poorly indurated sandstone that apparently has been extensively altered and remagnetized as a consequence of recent weathering.

Sites 5 and 6 represent the two extremes of demagnetization behavior (Fig. 3). Specimens from site 5, a dark-red fine-grained sandstone and siltstone, exhibit high stability to thermal demagnetization. About 90% of the NRM remains at 600°. Retention of such substantial proportions of NRM in red beds indicates that specular hematite carries most of the remanence [13,14]. Polished thin sections of samples form site 5 contain abundant specular hematite, some of which appears to be primary and detrital and some which exhibits secondary overgrowths on detrital specularite grains and replacements of biotite. In specimens from site 6, 40% of the NRM remains at 600°C. Rock at this site consists of pale red massive medium-grained quartzite in which only hematite pigment was observed. Because thermal demagnetization curves from other sites plot between the two curves shown in Fig. 3, the stable magnetizations at other sites presumably reside in various proportions of hematite pigment and specularite.

Vector demagnetization diagrams of four specimens display different characteristics (Fig. 4). The

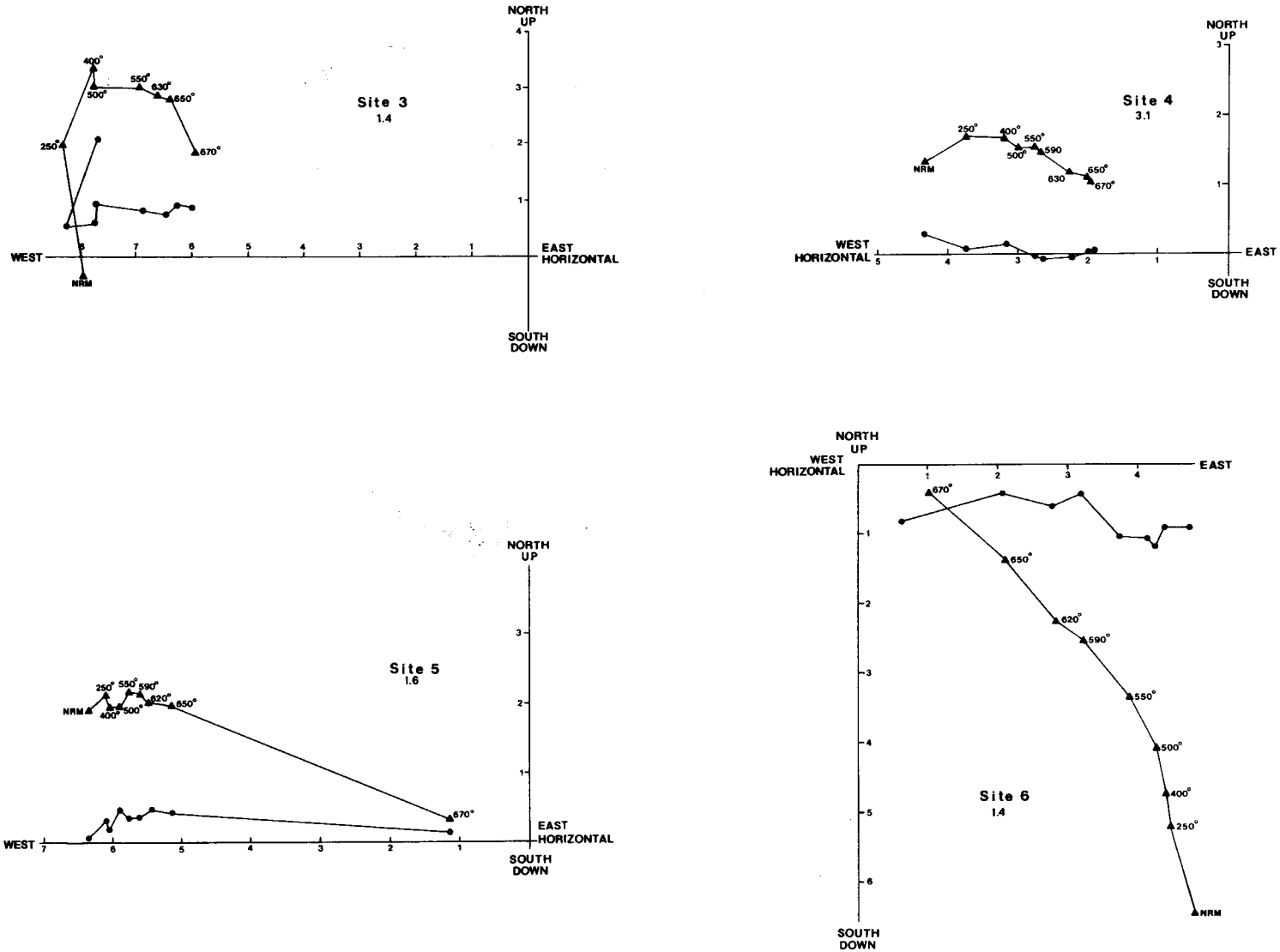


Fig. 4. Vector demagnetization diagrams [25]. The inclination paths (triangles) plot the vertical component against the horizontal. The declination path (circles) shows the horizontal projection. Temperature in °C. Numbers beneath site identification are sample numbers. 1 unit = 10^{-6} G.

TABLE 1

Site	Stratigraphic unit	Location	Lithology
6	Upper part of quartzite of Hades Pass	Moon Lake 40.580N 110.520W	pale red medium-grained quartzite
5	Lower part of quartzite of Hades Pass	Hayden Fork, Bear River 40.775N 110.883W	dark red medium to fine-grained sandstone and siltstone
4	Quartzite of Mount Watson	Hayden Fork, Bear River 40.752N 110.878W	pale grayish purple, fine- to medium-grained quartzite and arkose
3	Shale Moosehorn Lake	Bald Mountain Pass 40.687N 110.896W	dark reddish brown siltstone and shale, minor quartzite and arkose interbeds
2	Middle(?) part of Uinta Mountain Group	Crouse Creek 40.820N 109.090W	ref fine to coarse sandstone and quartzite
1	Lower part of Uinta Mountain Group	Cold Spring Mountain 40.992N 109.017W and Jesse Ewing Canyon 40.930N 109.180W	reddish brown, fine- to medium grained sandstone and quartzite, minor conglomerate and shale
(1–5)	Mean, Uinta Mountain Group		

curve for site 3 shows the removal of a low blocking temperature component (400°C) having a northerly declination and positive inclination, which is probably a viscous component of magnetization acquired in late Cenozoic time. The other three curves illustrate behavior more typical than that observed at site 3. Where present, declination changes are slight, and changes in inclination reflect removal of downward (positive) components of inclination, apparently reflecting removal of present field components of magnetization. No intermediate components of magnetization were observed in vector plots for any specimen.

4. Directions of magnetization

No conclusive evidence, such as conglomerate or fold tests, yet exists to demonstrate an early age for the magnetization. Therefore, paleomagnetic directions reported here are considered preliminary. Although petrographic observations suggest that much of the hematite pigment is predigenetic, they shed no light as to the amount of time required for hematite formation. Studies are planned to provide information on the time of acquisition of the magnetization.

Directions of magnetization after partial thermal demagnetization are plotted in Fig. 5 for samples from each site. The mean NRM and cleaned directions for each site are also plotted in Fig. 5 and are listed in Table 1. Westerly declinations, with corresponding pole positions in the Pacific Ocean, are considered to be normal polarity, following the convention employed for directions obtained from the Belt and Grand Canyon Supergroup [15].

Lower part of the Uinta Mountain Group (site 1). Samples from the lowermost part of the Uinta Mountain Group at Jesse Ewing Canyon (site 1A) and from the lower part of the Uinta Mountain Group at Cold Spring Mountain (site 1B) have been combined because of their stratigraphic and geographic proximity. Directions from 15 of 41 samples collected were discarded because they were

Polarity	Treatment	n	D (°)	I (°)	α_{95} (°)	k	Paleomagnetic pole		δ_m	δ_p
							lat.	long.		
R	NRM	8	91.7	72.8	22.6	6.9				
R	650°C	8	106.3	22.1	15.9	13.0	4.6N,	140.1E	16.8	8.9
N	NRM	48	275.3	−1.0	5.4	15.0				
N	550°C	48	267.2	−15.3	1.3	241.9	7.2S,	155.0E	1.4	0.7
R	NRM	6	77.7	−7.8	38.5	3.8				
R	550°C	6	93.6	−24.4	10.5	41.8	11.0N,	167.0E	11.2	6.0
Mean	550°C	2	270.3	4.5	–	–	1.5N,	160.6E		
N	NRM	7	284.3	5.5	27.7	5.7				
N	550°C	7	278.6	−3.8	26.0	6.3	5.3N,	152.0E	26.0	13.0
N	NRM	14	276.9	−1.8	4.2	91.9				
N	550°C	14	269.8	−13.1	4.4	83.5	4.5S,	154.2E	4.5	2.3
N	NRM	13	276.5	15.0	18.5	6.0				
N	620°C	13	274.0	0.1	9.6	19.8	3.1N,	158.3E	9.6	4.8
N	NRM	26	266.4	17.7	15.9	4.2				
N	590°C	26	267.8	−3.3	8.6	11.9	2.7S,	161.1E	8.6	4.3
					5.4	201.7	0.6N,	157.2E	~5.4	~2.7

on or near the present axial dipole field direction or were widely divergent from the main group. The remaining 26 sample directions are well grouped in a normal polarity direction (Fig. 5). A cleaning temperature of 590°C produced the best grouping of directions, although the mean directions were essentially identical for all temperatures between 500 and 650°C.

Middle(?) part of the Uinta Mountain Group (site 2). Samples were collected from three exposures encompassing about 400 m of section at Crouse Creek on the south flank of the eastern Uinta Mountains (site 2). The precise stratigraphic position is uncertain, but it probably is in the lower middle part of the 7-km-thick undivided section. Directions from 20 of 33 samples collected did not respond to demagnetization and lay on or near the present axial dipole field direction. The remaining 13 samples have a well-grouped normal polarity direction (Fig. 5). The 20 discarded samples came from friable coarse-grained sandstone, whereas the samples that were retained came from fine grained sandstone and siltstone. A temperature of 620°C was selected for the calculation of a mean direction because it resulted in the best grouping of directions.

Shale of Moosehorn Lake (site 3). Fourteen samples were obtained from a small outcropping of the shale of Moosehorn Lake along State Highway 150, near the crest of the western Uinta Mountains. Directions from all samples from this dark-red fine-grained sandstone and siltstone are of normal polarity and are well clustered (Fig. 5). Optimum grouping occurred at 550°C.

Quartzite of Mount Watson (site 4). Seven of ten samples obtained from the quartzite of Mount Watson along State Highway 150 at site 4 exhibit a normal polarity grouping following demagnetization at 550°C. Scatter of directions is greater in this unit than in any other (Fig. 5), probably because of the sparseness of hematite in the pale pink quartzite. Intensity of NRM for these samples is $\sim 1 \times 10^{-7}$ G compared to $\sim 1 \times 10^{-5}$ G for samples from the darker red units.

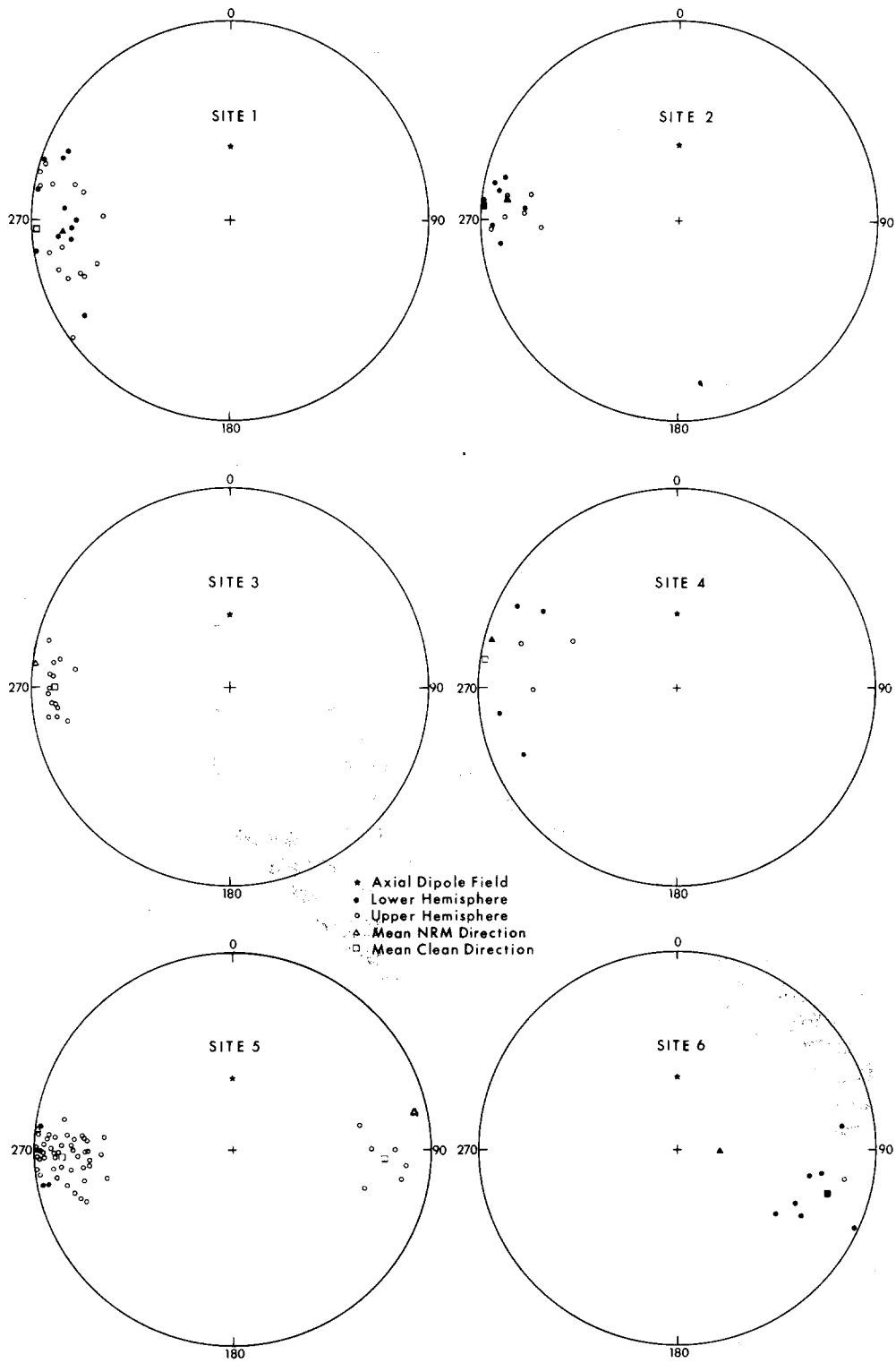


Fig. 5. Equal-area stereographic projection of selected demagnetized samples from six stratigraphic levels within Uinta Mountain Group. Mean directions for sample groups are listed in Table 1.

Lower part of the quartzite of Hades Pass (site 5). The lower part of the quartzite of Hades Pass was sampled along State Highway 150 on the north flank of the range (site 5). Both normal and reversed polarity directions are present in a narrow (4 m) stratigraphic interval, but the directions are not antiparallel ($D = 93.6°$, $I = -24.4°$, $\alpha_{95} = 10.5°$ for reversed-polarity samples; $D = 267.2°$, $I = -15.3°$, $\alpha_{95} = 1.3°$ for normal-polarity samples). The reversed-polarity directions came from six samples collected from horizons from which normally magnetized samples also were obtained. The non-interbedded stratigraphic distributions and the negative inclinations of both polarities suggest the presence of a stable, negative-inclination, secondary component of magnetization. This inference is supported by eight samples, not shown on Fig. 5, which have directions between those of the normal and reversed endpoints. Because directions from the eight samples, which probably contain nearly equal amounts of both components of magnetization, did not shift significantly during demagnetization, blocking temperature spectra are probably similar for both components of magnetization. Secondary specular hematite was seen in polished thin sections, which probably is the principal source of this stable secondary magnetization. An approximate primary direction of magnetization can be obtained by averaging the normal direction and the anti-pole of the reversed-polarity mean direction. Although statistics cannot be obtained from such an average, the resulting direction probably is more nearly correct than either the normal- or reversed-polarity mean direction.

Upper part of the quartzite of Hades Pass (site 6). Thirteen samples were collected from the upper part of the quartzite of Hades Pass at Moon Lake on the south flank of the Uinta Mountains (site 6). Directions from eight samples, after demagnetization at 650°C, grouped reasonably well in a reverse-polarity direction. The other five samples had widely scattered directions. This group of samples, in which no specularite was observed, showed the greatest change in direction upon demagnetization of all sample groups studied (Fig. 5 and Table 1), attesting to the lower magnetic stability of hematite pigment.

5. Paleomagnetic poles

Paleomagnetic poles have been calculated from the mean cleaned directions from each site (Table 1), based on the assumption that sufficient time is represented at each site to have averaged out the secular variation. Poles and their 95% confidence ovals are plotted in Fig. 6. Poles for most sites lie near each other, in the equatorial Pacific Ocean. No systematic apparent polar motion is exhibited within the Uinta Mountain Group, with the possible exception of the pole from the stratigraphically highest site sampled, the upper part of the quartzite of Hades Pass (site 6). This pole (pole 6, Fig. 6) lies west of the other poles. However, because the 95% confidence ovals overlap and because the pole is based on only eight samples, verification of the apparent shift awaits additional sampling.

In light of the clustering of poles, a mean pole has been calculated using poles from all sites except site 6. This mean pole is listed in Table 1 and is plotted with the pole from site 6 in Fig. 7.

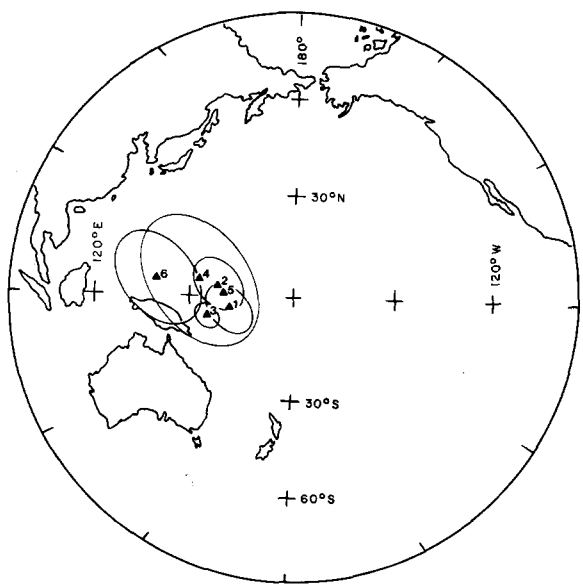


Fig. 6. Plot of paleomagnetic poles for Uinta Mountain Group and their 95% confidence ovals. Pole numbers are listed in Table 1. No oval is plotted for pole 5, as it is calculated from mean normal and reversed directions.

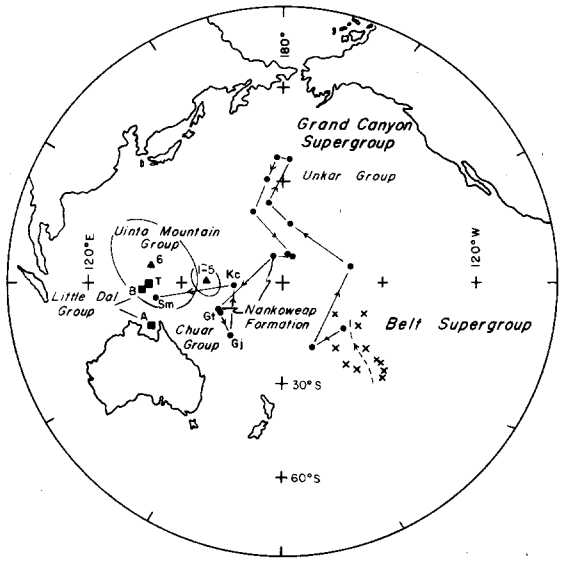


Fig. 7. Plot of a mean paleomagnetic pole (from sites 1–5) for Uinta Mountain Group and a pole (6) form the stratigraphically highest site. Ovals represent 95% confidence zone about the poles. Apparent polar wander paths from Belt Supergroup [15] (×'s) and Grand Canyon Supergroup [10,16] (circles) shown. Poles of Chuar Group are labeled as follows: *Gt*= Tanner Member of Galeros Formation, *Gj*=Jupiter Member of Galeros Formation, *Kc*=Carbon Butte Member of Kwagunt Formation, and *Sm*=Sixtymile Formation. All from Elston and Bressler [16]. Poles *A* and *B* from Little Dal Group of Canada [17,18] (squares) are detrital and chemical components, respectively. Pole *T* is from Tsezotone sills, which intrude Little Dal Group [19].

Generalized apparent polar wandering paths from two sequences that have been considered to be correlative with the Uinta Mountain Group are shown in Fig. 7. The two sequences are the Belt Supergroup of western Montana [15] and the Grand Canyon Supergroup of Arizona [10,16]. Eleven poles obtained across a substantial part of the Belt Supergroup control its path, and poles form 19 stratigraphic units control the Grand Canyon polar path. Also shown in Fig. 7 are three poles recently reported from the Little Dal Group of the Mackenzie Mountains, Northwest Territories, Canada[17–19].

Elston and Bressler [15,16] suggest on paleomagnetic grounds that the Belt Supergroup is largely, if not entirely, older than the Grand Canyon Supergroup. Elston and Grommé [10] have evaluated the age of the Unkar Group of the Grand Canyon Supergroup based on a detailed correlation of the polar path and polarity zonation of the Unkar Group with the Keweenawan Supergroup of the Lake Superior region. The large north-trending loop in the pole path was formed in the interval from about 1200 to 1100 m.y. ago, based on isotopic ages form Arizona and Lake Superior that accord with the paleomagnetic data. The Nankoweap Formation of the Grand Canyon Supergroup unconformably overlies the 1090 ± 70-m.y.-old Cardenas Lavas [20] at the top of the Unkar Group. The Nankoweap itself contains an unconformity that separate its lower and upper members, and an unconformity separates the Nankoweap Formation from the overlying Chuar Group. An age of ~ 1050 m.y. therefore, is tentatively assigned to the Nankoweap Formation [16]. The Chuar Group is nearly 1.9 km thick and consists of the Galeros and Kwagunt Formations, overlain by Sixtymile Formation, which is 60 m thick [21,22]. The Chuar Group also contains the fossil acritarch *Chuaria*, although its size differs somewhat from specimens found in the Red Pine Shale of the Uinta Mountain Group [11].

The prolonged episode of quiet marine deposition that produced the Chuar Group was brought to a close by the Grand Canyon orogeny [21]. Uplift and faulting during this event produced landslide breccia deposits within the overlying Sixtymile Formation, and the uplift is considered by Elston and McKee [22] to be reflected in a partially(?) reset K-Ar age of about 930 m.y. from the mineral analysis of diabase sills in the Unkar Group and from the older crystalline basement, and a more nearly completely reset age of 825 m.y. from whole rock analysis of samples of the Cardenas Lavas at the top of the Unkar Group. They propose that the 825-m.y. reset age more closely records the time of uplift and faulting of the Grand Canyon orogeny.

The age of the Little Dal Group of Canada, which contains *Chuaria* [23] was inferred by Park [17,18] to be 950 m.y. However, an internal Rb-Sr isochron age for two sills that were intruded shortly after deposition of the Little Dal Group is reported to be 770 ± 20 m.y. [24].

Poles for the Uinta Mountain Group are dis-

tinctly separate from poles of the Belt Supergroup and the Unkar Group of the Grand Canyon Supergroup, but lie near poles from the upper member of the Nankoweap Formation and the Chuar Group of the Grand Canyon supergroup. The westerly shift of the pole from the Chuar poles to the poles for the Sixtymile Formation and the Little Dal Group appears also to occur at the top of the Uinta Mountain Group (site 6), although this apparent shift remains to be verified.

If an age range for the upper member of the Nankoweap Formation and the Chuar Group of about 1050 to 825 m.y. is accepted, the paleomagnetic correlation implies a similar age range for the Uinta Mountain Group. The presence of *Chuaria*, the overlap of isotopic ages, an 1100-m.y. age for the base of the Uinta Mountain Group, and the general agreement of poles with those of the Little Dal Group, which also contains *Chuaria* and which was deposited before 770 m.y. ago, further support the paleomagnetic correlation and age assignment for the Uinta Mountain Group.

Acknowledgements

I wish to thank C.A. Wallace and W.R. Hansen for providing helpful information regarding the stratigraphy of the Uinta Mountain Group, and for suggesting sampling locations. The enthusiastic assistance of M.E. Ahkeah, D.P. Elston and A.L. Luna in the field is much appreciated. I am grateful to D.P. Elston for providing data from the Grand Canyon Supergroup. Suggestions of D.P. Elston, J.T. Hagstrum, J. Hillhouse, and E. Mankinen greatly improved the original manuscript.

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